Robotic Exoskeleton for Load Transportation

IMU-based locomotion intent recognition and assist gating in MATLAB

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ExoTechHK Limited (HKSTP) • Final Year Project (MAEG4999)

Key message: the validated baseline is a reproducible HuGaDB MATLAB assist-gating pipeline: 30-D IMU features feed a binary active-vs-inactive SVM, and FSM hysteresis turns noisy window predictions into a stable assist on/off command.

Binary SVM
5-fold OOF
USC-HAD 99.44%
HuGaDB 96.80%
Benchmark comparison across datasets

Multiclass SVM
5-fold OOF
USC 53.41%
HuGaDB 75.78%
Native multiclass activity recognition per dataset

Sequence baseline
Binary LSTM, 20% holdout
USC-HAD 99.68%
HuGaDB 98.36%
Separate holdout protocol, supporting baseline

GitHub IG

Problem and approach

- Manual load handling above 20 kg raises fatigue and musculoskeletal injury risk.
- An exoskeleton is useful when assistance engages and releases at the right time.
- This work targets a **software supervisor**: infer active movement from IMUs and issue a reliable assist on/off signal without EMG.
- The goal is not full torque control yet; it is a robust decision layer that can sit above future hardware control.

Implemented workflow

- **Input:** HuGaDB for training and held-out HuGaDB replay for simulation; USC-HAD is retained as a comparison benchmark.
- **Fusion:** Kalman filtering is optional and used for a replay-time segment-pitch visualization trace, not for classifier preprocessing.
- **Windowing:** 1.0 s windows at 100 Hz with 50% overlap.
- **Features: 30-D** handcrafted vector, five statistics per IMU slot across six slots.
- **Primary model:** RBF SVM for **binary gating**: active-movement windows vs. inactive windows, using dataset-specific activity-to-binary maps.
- **Supervisory smoothing:** FSM hysteresis, 3 active predictions to engage assist and 5 inactive predictions to release.

Main validated result

The strongest binary-SVM claim in the current repository is the **HuGaDB** model: it reaches **96.80%** five-fold out-of-fold accuracy on the HuGaDB benchmark split.

Training data	OOF acc.
USC-HAD	**99.44%**
HuGaDB	**96.80%**

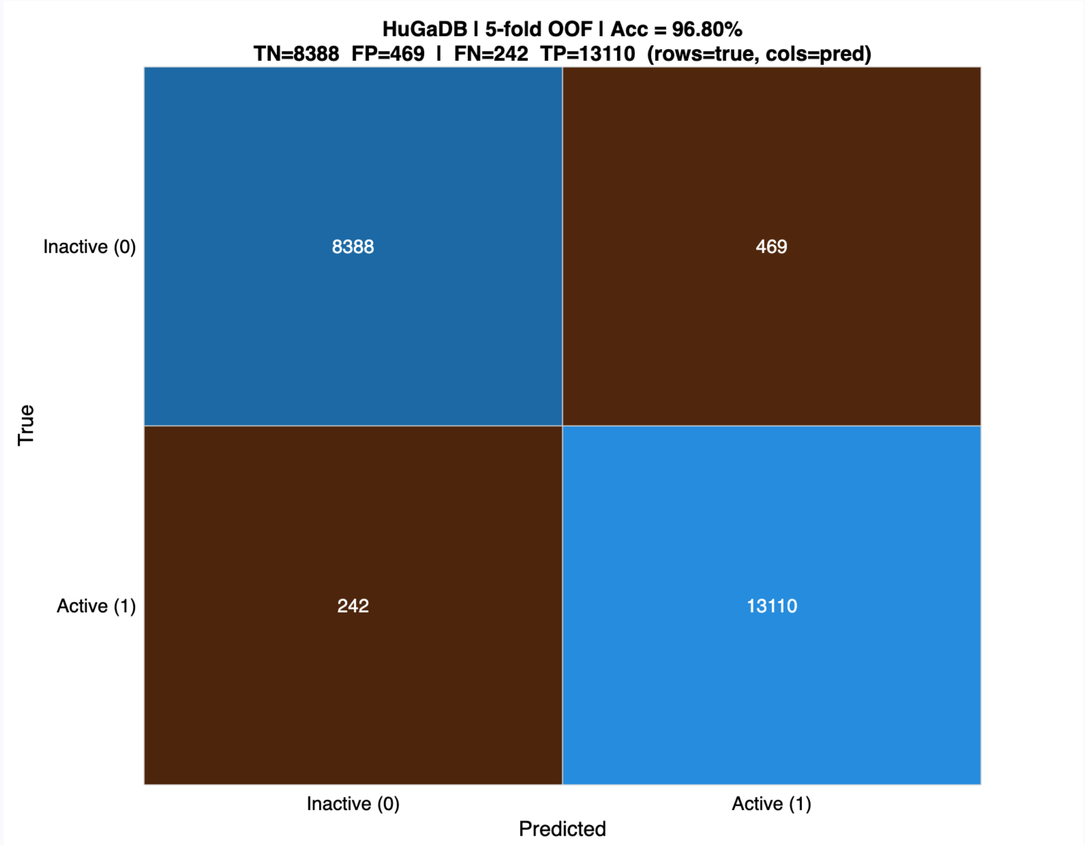
This table is kept as **benchmark context**. The final project path does **not** merge datasets; it trains on HuGaDB and keeps replay-based simulation as a separate step.

Data and evaluation protocol

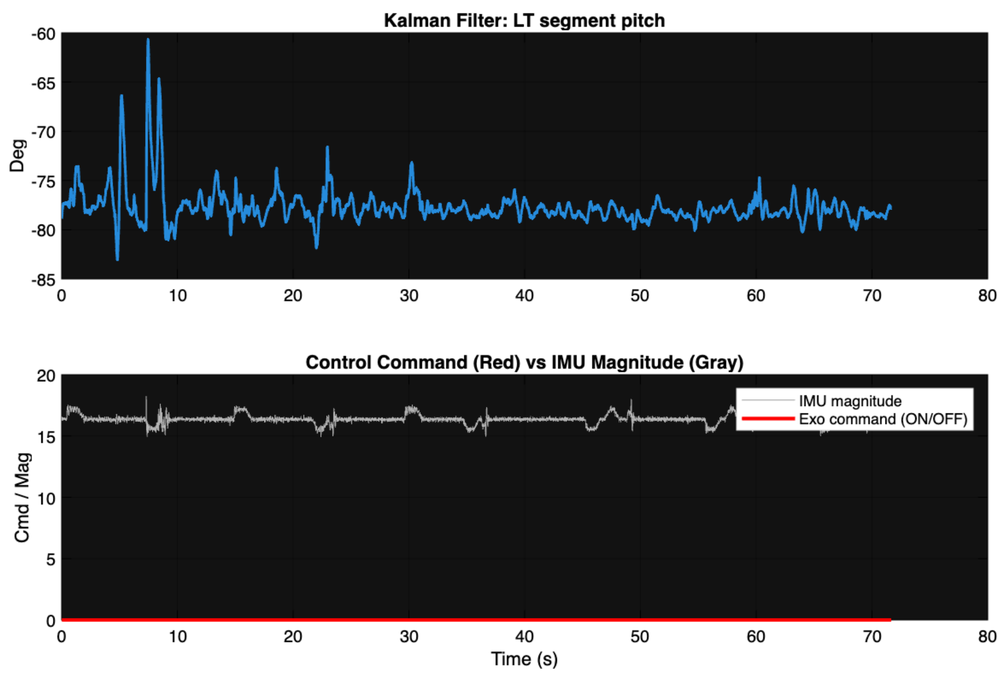
- **HuGaDB** contributes six on-body IMU streams per window and is the dataset used by the final pipeline.
- **USC-HAD** contributes one IMU stream; it remains in the poster to compare dataset effects under the same 30-D layout.
- The **binary SVM** is the main baseline; the active checkpoint is the HuGaDB model.
- **Binary** asks a simple control question: should assist stay **off** or turn **on**? **Multiclass** asks a harder recognition question: which exact activity is occurring?
- **Multiclass SVM** (12 USC / 12 HuGaDB classes) and **binary/multiclass LSTM** are supporting baselines; the SVM uses stratified subsamples and the LSTMs use separate stratified 20% holdouts, so neither is a direct deployment comparison.

Pipeline at a glance

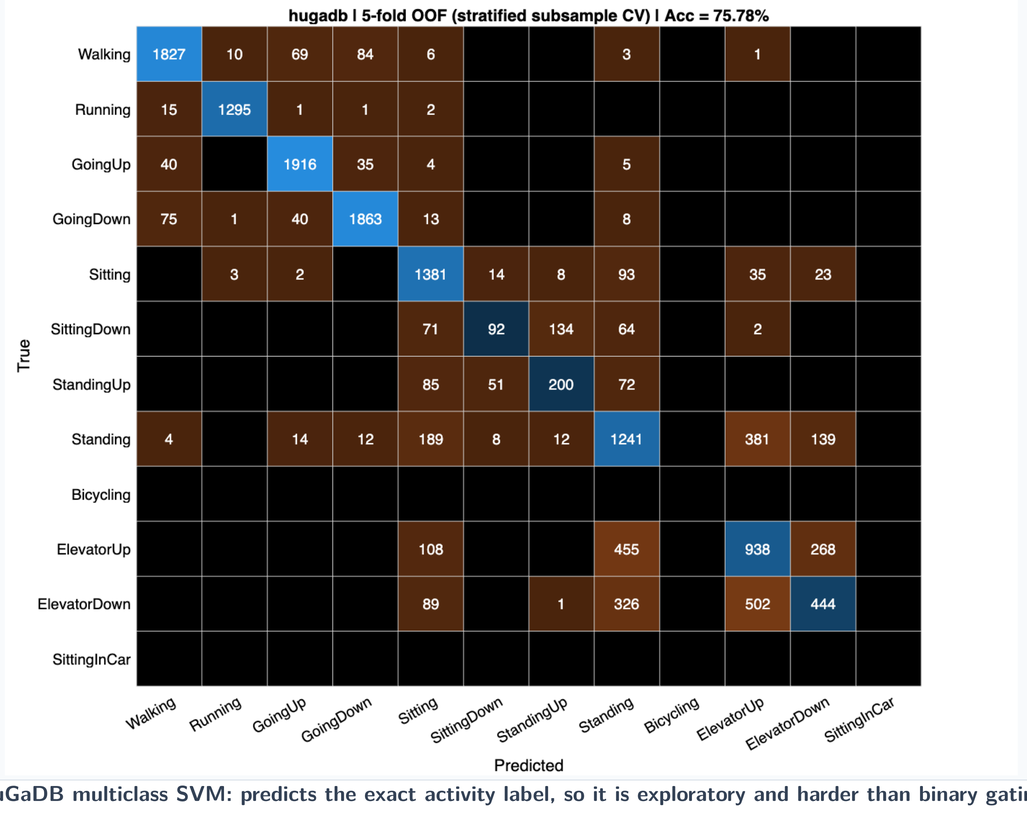
1. HuGaDB train pool and held-out replay → 2. Optional Kalman visualization trace → 3. 30-D features per window → 4. Binary SVM prediction → 5. FSM hysteresis for assist → 6. Stable on/off command



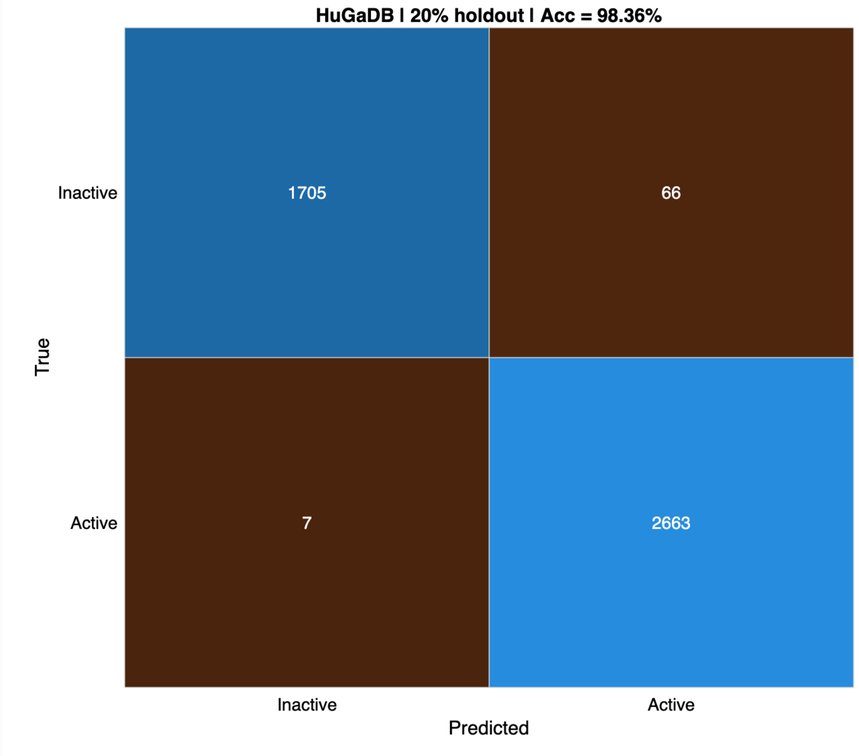
HuGaDB binary SVM: predicts assist-gating on/off, not the exact activity class



Held-out HuGaDB replay: Kalman visualization plus FSM-latched assist output



HuGaDB multiclass SVM: predicts the exact activity label, so it is exploratory and harder than binary gating



Binary LSTM: supporting sequence baseline on a separate 20% holdout split

Take-home message: the repository now delivers a traceable MATLAB baseline for IMU-based binary assist gating, centered on HuGaDB training, benchmark comparison tables, replay-based simulation, and FSM-stabilized control output.

How to read the supporting baselines

- The pipeline plot shows why the FSM matters: it turns noisy window-level predictions into a smoother and more usable assist signal.
- The **binary** task is easier because several activities are merged into two groups; the **multiclass** task is harder because it must separate the exact activity label.
- Multiclass SVM OOF scores (**53.41%** USC-HAD; **75.78%** HuGaDB) are informative for analysis, but not yet strong enough for fine-grained control.
- Binary LSTM holdout accuracy is **99.68%** on USC-HAD and **98.36%** on HuGaDB; it uses a different validation protocol and remains a supporting baseline rather than a direct SVM comparison.
- Multiclass LSTM holdout accuracy is **91.10%** on USC-HAD and **83.80%** on HuGaDB; because the datasets remain distinct native label spaces, the comparison is a summary metric.

Current limits

- No hardware-in-the-loop or human-subject load trials are reported here.
- The 100–200 ms predictive horizon remains a **design target**, not a re-benchmarked final result.
- Current output is **assist gating**, not full torque scheduling or terrain-aware control.
- Multiclass and LSTM baselines still need a stronger evaluation protocol aligned with the main binary SVM path.

Next steps

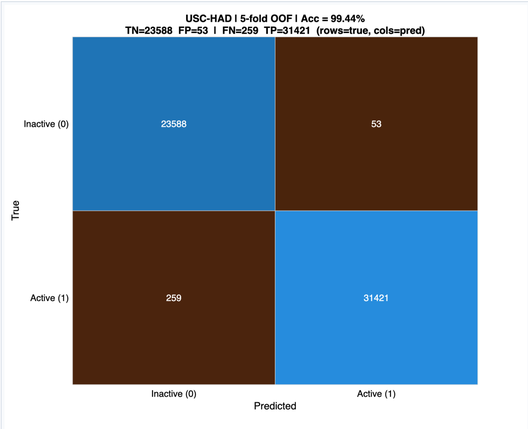
- Align multiclass and LSTM evaluation with the same level of rigor used for the main binary SVM baseline.
- Add hardware integration, latency measurement, and load-carrying trials.
- Move from binary assist gating toward subject-specific, activity-aware assistance.

Summary

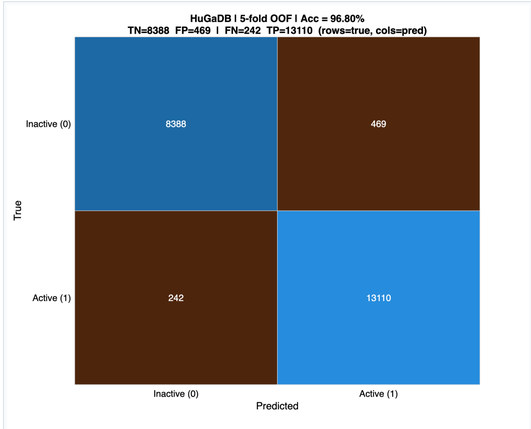
- **Delivered:** HuGaDB binary SVM, optional Kalman visualization, FSM smoothing, reproducible metrics, and replay in MATLAB.
- **Best validated result:** HuGaDB binary SVM achieves **96.80%** five-fold OOF accuracy.
- **System role:** simulation-level assist gating supervisor.
- **Value:** a credible, traceable software baseline that separates benchmark comparisons from the final deployed method.

Complete Generated Figure Gallery

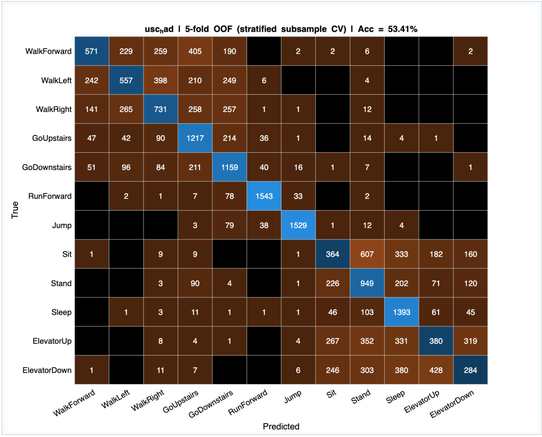
All distinct PNG artifacts currently written under `results/figures/`.



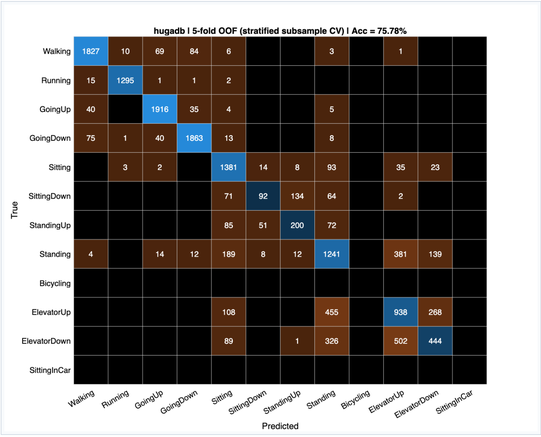
Binary SVM
USC-HAD



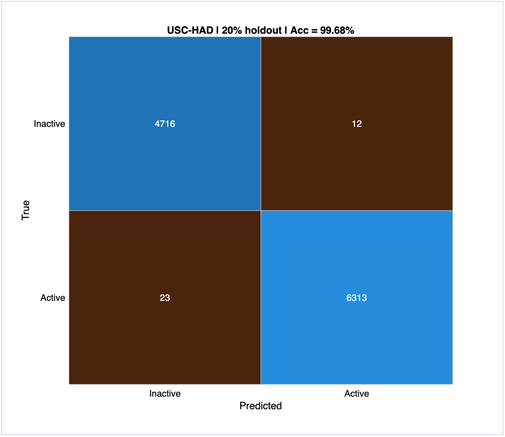
Binary SVM
HuGaDB



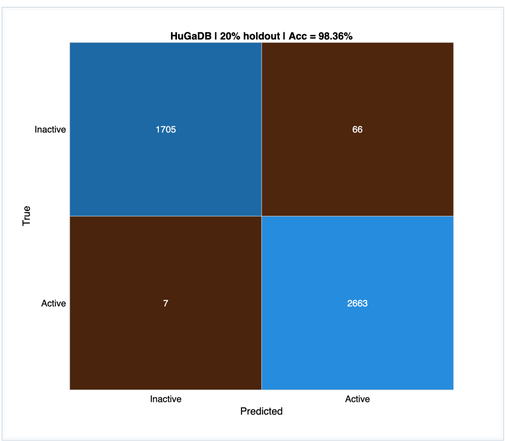
Multiclass SVM
USC-HAD



Multiclass SVM
HuGaDB



Binary LSTM
USC-HAD



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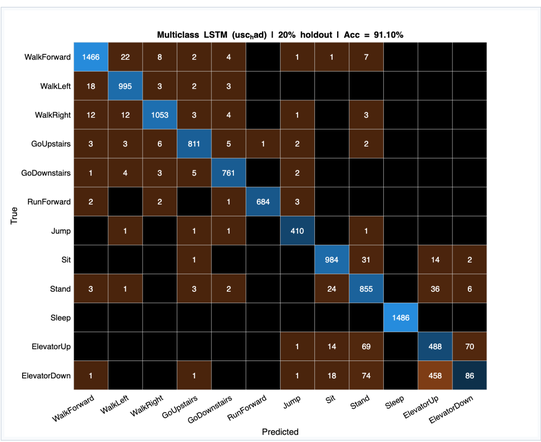
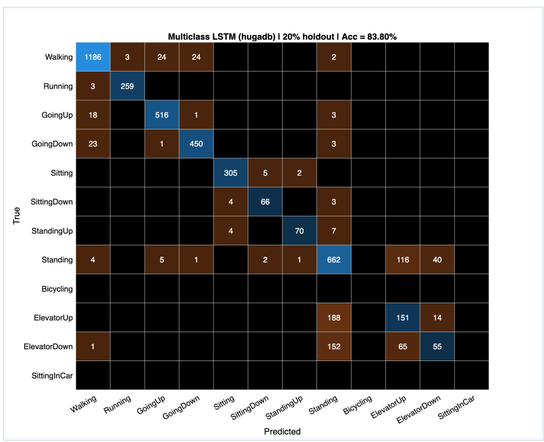
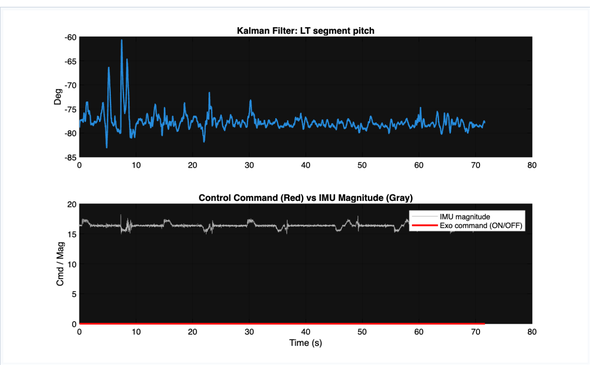


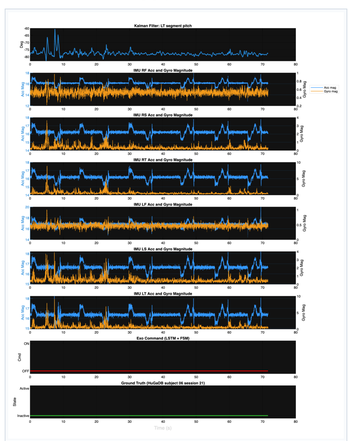
Multiclass LSTM
USC-HAD



Multiclass LSTM
HuGaDB



Binary SVM
pipeline replay



Binary LSTM
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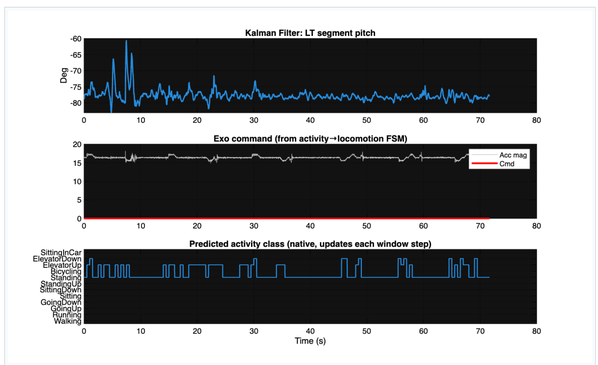

Multiclass SVM
pipeline replay